

Methodological Framework Support

Project: Enhancing Urban Resilience through Green Infrastructure (Kathmandu Case Study)

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1. Purpose and Role of the Methodological Framework Support

The purpose of this section is to establish a clear scientific bridge between the reviewed literature and the practical methodology adopted in this project. While the literature review synthesizes what previous studies have demonstrated regarding urban flooding, green infrastructure performance, and sediment dynamics, the methodological framework defines how these findings are operationalized within the project's simulations, structural designs, and analytical workflows.

This support section ensures that all modeling decisions are:

- Evidence-based, not arbitrary;
- Consistent with regional and international standards;
- Transparent and reproducible, fulfilling Erasmus Mundus evaluation criteria.

Furthermore, this framework enables interdisciplinary coordination by translating literature-derived parameters into actionable inputs for hydrological analysis, computational modeling, and structural design.

2. Preliminary Validation of Methodological Choices

To demonstrate methodological rigor, each major modeling and design decision is validated against peer-reviewed literature, institutional guidelines, or internationally recognized engineering standards. The validation table (provided separately in tabular format) summarizes this alignment; the underlying logic is elaborated below.

2.1 Selection of Hydrological Method

The Rational Method ($Q = CiA$) is adopted as the primary hydrological approach for runoff estimation. This method is widely recommended for small to medium urban catchments ($< 50 \text{ km}^2$) and remains a standard approach within South Asian urban engineering practice.

Nepal Road Standards and Department of Hydrology and Meteorology (DHM) guidelines endorse the Rational Method for preliminary drainage and flood estimation in highly urbanized areas. Given the fragmented and small-scale nature of Kathmandu's urban sub-catchments, this method provides a suitable balance between simplicity, transparency, and engineering relevance.

The method also allows direct integration of imperviousness and rainfall intensity parameters derived from literature, making it ideal for “before vs after” green infrastructure comparison scenarios.

2.2 Impervious Surface Representation

An impervious surface ratio of 85% is assumed for core urban areas of Kathmandu Valley. This assumption is grounded in urban land-use studies that document rapid conversion of permeable surfaces into roads, rooftops, and paved areas due to unplanned urban growth.

- This high imperviousness assumption serves two methodological purposes:
- It represents a conservative, worst-case baseline for flood modeling.
- It ensures that the effectiveness of green infrastructure interventions is evaluated against realistic urban conditions rather than idealized scenarios.

2.3 Rainfall Intensity and Flood Threshold Definition

Rainfall intensity is a critical driver of urban flooding in monsoon-dominated regions. Based on DHM rainfall records and IPCC (2022) assessments of climate extremes in the Himalayan region, the project adopts a 10-year return period rainfall intensity of 140 mm/hr.

This value reflects:

- Increasing frequency of short-duration, high-intensity rainfall events.
- Observed exceedance of traditional drainage design capacities during monsoon storms.
- The need to test system resilience under climate-stressed conditions, rather than average rainfall events.

Additionally, a flood depth threshold of 0.30 m is selected to classify high-risk flooding conditions. International urban flood safety literature identifies this depth as the point beyond which pedestrian mobility, vehicular movement, and emergency access become severely compromised. This threshold therefore provides a meaningful indicator for impact-based flood assessment rather than purely hydraulic exceedance.

2.4 Green Infrastructure Effectiveness Parameters

Green infrastructure (GI) performance parameters are derived primarily from ADB (2022) technical guidance and regional NBS literature. A 40% runoff reduction is adopted as a conservative effectiveness estimate, positioned within the reported 30–60% range for permeable pavements and vegetated systems in tropical climates.

This conservative selection accounts for:

- High rainfall intensities that may exceed infiltration capacity.
- Silt-rich soils prevalent in Kathmandu Valley.
- Partial performance loss due to maintenance constraints.

For bio-swales, infiltration rates of 2.5–4.0 cm/hr are adopted based on reported values from South Asian case studies. These rates are particularly relevant for modeling infiltration-driven runoff attenuation under monsoon rainfall conditions.

2.5 Sedimentation and Maintenance Considerations

Sedimentation significantly influences long-term drainage and GI performance in Kathmandu. Construction activities, upstream erosion, and unpaved surfaces contribute to high sediment loads during rainfall events.

Based on ADB technical guidance, the project assumes:

Up to 70% sediment trapping efficiency for vegetated GI systems.

Approximately 15% annual reduction in infiltration capacity in the absence of routine maintenance.

Including sedimentation explicitly in the methodological framework prevents overestimation of GI performance and enhances the realism of long-term flood mitigation scenarios.

3. Justification of Key Modeling Assumptions

3.1 Assumption Transparency

All assumptions are explicitly stated and justified to avoid implicit or hidden modeling biases. This transparency is essential for both academic rigor and interdisciplinary coordination.

3.2 Data Limitations and Mitigation Strategy

A primary limitation of the study is the lack of localized, high-resolution (hourly) drainage sensor data within inner Kathmandu. To mitigate this limitation:

Secondary rainfall and hydrological data from DHM are utilized.

A 10% safety margin is applied to runoff and flood depth outputs.

Conservative parameter values are consistently favored over optimistic estimates.

This approach aligns with best practices in climate-resilient infrastructure modeling and ensures that results remain robust under uncertainty.

4. Integration with Team Methodology

The validated parameters derived through this framework are explicitly transferred to other team members to maintain consistency across disciplines.

Structural Design Interface (Santosh Aryal)

Hydrological outputs—such as runoff coefficients, peak discharge, and flood depth thresholds—inform hydraulic loading and flow conditions required for climate-smart structural design in AutoCAD Civil 3D.

Computational Modeling Interface (Sudeep Rauniyar)

Rainfall intensity, infiltration rates, and GI effectiveness values are directly embedded into Python-based simulations and ANSYS Fluent models to generate comparative pre- and post-intervention flood scenarios.

This coordinated handover ensures that all analytical components of the project are built upon a shared, literature-validated foundation.

5. Contribution to Phase 1 and Beyond

This Methodological Framework Support:

- Strengthens the scientific credibility of Phase 1 outputs.
- Demonstrates rigorous parameter validation and assumption management.
- Lays a solid foundation for Phase 2 modeling, comparative analysis, and publication.